Chapter 3

Complex Integration

Definition 3.1. The *primitive of a function* $f : \mathcal{D} \to \mathbb{C}$ is an analytic function $F : \mathcal{D} \to \mathbb{C}$ such that $F' = f$ for any $z \in \mathcal{D}$.

Motivation 3.2. In the case of $\mathcal{D} \subseteq \mathbb{R}$, the first fundamental theorem of calculus (Theorem B.127) guarantees the existence of the primitive of any function $f : \mathcal{D} \to \mathbb{R}$, since we can construct a suitable integral as the primitive of f. However, this is not true in the case of $\mathcal{D}$ being an open subset of $\mathbb{C}$: a function that has a primitive must be analytic, c.f. Theorem 3.66. Therefore, it is natural to ask the question whether the primitive of a function $f : \mathcal{D} \to \mathbb{C}$ exists; this question is a key problem in our extension of $\mathbb{R}$ to $\mathbb{C}$.

Roadmap 3.3. This chapter mainly concerns the equivalence of the following four statements.

(a) f has a primitive.

(b) The complex integral of f on a closed curve vanishes.

(c) The complex line integral of f is path independent.

(d) f is analytic.

After examining a stepping-stone integral in Definition 3.4, we formally define the concept of complex line integrals in Section 3.2 and prove Theorem 3.16 that states (a) $\Rightarrow$ (b). One key result of Section 3.3 is Theorem 3.37, which states that (a) $\Leftrightarrow$ (b) $\Leftrightarrow$ (c). The central result of Section 3.3 is the Cauchy theorem, which states (d) $\Rightarrow$ (b) in Theorems 3.40, 3.47, 3.51, 3.54, and 3.57 on various domains. In Section 3.4, the generalized Cauchy integral formula (Theorem 3.66) gives (a) $\Rightarrow$ (d); this is utilized in the proof of the Morera theorem 3.69, which states (b) $\Rightarrow$ (d).

3.1 A stepping stone

Definition 3.4. A function $f : [a, b] \to \mathbb{C}$ with $a, b \in \mathbb{R}$ is called *integrable* iff $\operatorname{Re} f, \operatorname{Im} f : [a, b] \to \mathbb{R}$ are Riemann integrable as in Definition B.106; then

$$\int_a^b f(x)\mathrm{d}x = \int_a^b \operatorname{Re} f(x)\mathrm{d}x + \mathbf{i}\int_a^b \operatorname{Im} f(x)\mathrm{d}x. \tag{3.1}$$

Corollary 3.5. Denote by $\mathcal{V}$ the set of continuous functions $[a, b] \to \mathbb{C}$. Then $\mathcal{V}$ is a vector space and the integral $\int_a^b$ defined in (3.1) is a linear functional on $\mathcal{V}$.

Proof. We need to show that the integration is a linear map from $\mathcal{V}$ to $\mathbb{C}$, which holds because

$$\int_a^b (f(x) + g(x))\,\mathrm{d}x = \int_a^b f(x)\mathrm{d}x + \int_a^b g(x)\mathrm{d}x;$$
$$\forall \lambda \in \mathbb{C},\ \int_a^b \lambda f(x)\mathrm{d}x = \lambda \int_a^b f(x)\mathrm{d}x.$$ □

Lemma 3.6. If $f : [a, b] \to \mathbb{C}$ is continuous and F is a primitive of f, i.e. $F' = f$, then

$$\int_a^b f(x)\mathrm{d}x = F(b) - F(a). \tag{3.2}$$

Proof. This follows from (3.1) and the second fundamental theorem of calculus (Theorem B.128). □

Lemma 3.7. If $f : [a, b] \to \mathbb{C}$ is integrable and satisfies $|f(x)| \le C$ for all $x \in [a, b]$, then we have

$$\left|\int_a^b f(x)\mathrm{d}x\right| \le \int_a^b |f(x)|\,\mathrm{d}x \le (b - a)C. \tag{3.3}$$

Proof. The second inequality follows from the integral mean value theorem B.124. To prove the first inequality, define $\int_a^b f(x)\mathrm{d}x = re^{\mathbf{i}\theta}$ and we have

$$\begin{aligned}\left|\int_a^b f(x)\mathrm{d}x\right| &= r = e^{-\mathbf{i}\theta}\int_a^b f(x)\mathrm{d}x = \int_a^b e^{-\mathbf{i}\theta} f(x)\mathrm{d}x \\ &= \operatorname{Re}\int_a^b e^{-\mathbf{i}\theta} f(x)\mathrm{d}x = \int_a^b \operatorname{Re}\left[e^{-\mathbf{i}\theta} f(x)\right]\mathrm{d}x \\ &\le \int_a^b \left|e^{-\mathbf{i}\theta} f(x)\right|\mathrm{d}x = \int_a^b |f(x)|\,\mathrm{d}x,\end{aligned}$$

where all steps follows from complex arithmetic. □

Example 3.8 (Lemma 3.7 as a special case of the Minkowski inequality in L^p space)**.** The first inequality in (3.3) is a direct consequence of the Minkowski inequality (Theorem F.11) on the L^p space,

$$\forall p \ge 1, \qquad \|f + g\|_p \le \|f\|_p + \|g\|_p, \tag{3.4}$$

where $f, g \in L^p(\Omega)$ and

$$\|f\|_p = \left(\int_\Omega |f(x)|^p \mathrm{d}x\right)^{\frac{1}{p}}. \tag{3.5}$$

This inequality is important in establishing that $(L^p, \|\cdot\|_p)$ is a normed linear space for $p \geq 1$. For $p < 1$, (3.4) holds for non-negative functions f, g with the inequality reversed:

$$\forall p < 1, \qquad \|f+g\|_p \geq \|f\|_p + \|g\|_p. \tag{3.6}$$

Then Lemma 3.7 follows from setting $f = u^2$, $g = v^2$, $p = \frac{1}{2}$ in (3.6) and using the inequality $\int_a^b u \leq \int_a^b |u|$.

Theorem 3.9. If $f : [a, b] \to \mathbb{R}^k$ is integrable, then $\|f\|$ is integrable and

$$\left\|\int_a^b f \mathrm{d}x\right\| \leq \int_a^b \|f\| \mathrm{d}x, \tag{3.7}$$

where $\|\cdot\|$ denotes the Euclidean norm in Definition C.176.

Proof. Write $f(x) := (f_1(x), \ldots, f_k(x))$, $y_i := \int_a^b f_i(x)\mathrm{d}x$, and we have $y := \int_a^b f(x)\mathrm{d}x = (y_1, ..., y_k)$ and

$$\|y\|^2 = \sum y_j^2 = \sum y_j \int f_j \mathrm{d}x. \tag{3.8}$$

For a fixed $x \in [a, b]$, the Cauchy–Schwarz inequality yields

$$\langle y, f(x)\rangle = \sum y_i f_i(x) \leq \|y\|\|f(x)\|.$$

Integrate the above inequality, apply (3.8), and we have

$$\|y\|^2 = \sum y_j \int_a^b f_j \mathrm{d}x \leq \|y\| \int_a^b \|f(x)\| \mathrm{d}x.$$

If $\|y\| \neq 0$, we divide both sides of the above inequality by $\|y\|$ to obtain (3.7). If $\|y\| = 0$, each y_i must be zero, hence (3.7) also holds trivially. □

Lemma 3.10. For continuously differentiable functions $u, v : [a, b] \to \mathbb{C}$, we have

$$\int_a^b u(x)v'(x)\mathrm{d}x = (uv)\big|_a^b - \int_a^b u'(x)v(x)\mathrm{d}x. \tag{3.9}$$

Proof. This follows from Lemma 3.6 and the product rule $(uv)' = uv' + u'v$ in Theorem 2.20. □

Lemma 3.11. Let $I_1, I_2 \subset \mathbb{R}$ be intervals, $a, b \in I_1$ and $\varphi : I_1 \to I_2$ continuously differentiable and $f : I_2 \to \mathbb{C}$ continuous. Then

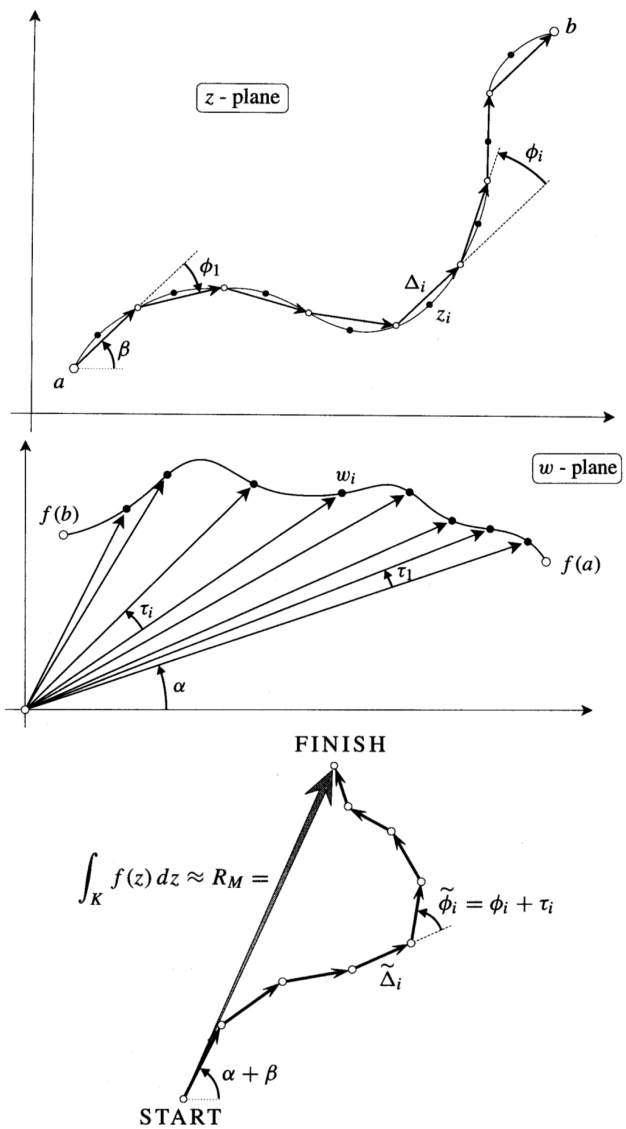
$$\int_{\varphi(a)}^{\varphi(b)} f(y)\mathrm{d}y = \int_a^b f(\varphi(x))\varphi'(x)\mathrm{d}x. \tag{3.10}$$

Proof. This follows directly from Theorem B.130 and Definition 3.4. □

3.2 Complex line integrals

Definition 3.12. For a piecewise smooth curve $K : [a, b] \to \mathbb{C}$ and a continuous function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \supset K([a, b])$, the *complex line integral* or *complex contour integral* of f along K, written $\int_K f$, is given by

$$\int_K f(z)\mathrm{d}z := \int_a^b f(K(t))K'(t)\mathrm{d}t. \tag{3.11}$$



Lemma 3.13. Let $\alpha : [c, d] \to \mathbb{C}$ be a piecewise smooth curve and $f : \mathcal{D} \to \mathbb{C}$ be a continuous function with $\alpha([c, d]) \subset \mathcal{D} \subset \mathbb{C}$. Let $\varphi : [a, b] \to [c, d]$ be a continuously differentiable function with $\varphi(a) = c$ and $\varphi(b) = d$. Then we have

$$\int_\alpha f(\zeta)\mathrm{d}\zeta = \int_{\alpha\circ\varphi} f(\zeta)\mathrm{d}\zeta. \tag{3.12}$$

Proof. Definition 3.12 yields

$$\begin{aligned} & \textstyle\int_\alpha f(\zeta)\mathrm{d}\zeta = \int_c^d f(\alpha(t))\alpha'(t)\mathrm{d}t \\ = \; & \textstyle\int_c^d f(\alpha(\varphi(s)))\alpha'(\varphi(s))\mathrm{d}\varphi(s) \\ = \; & \textstyle\int_a^b f[(\alpha\circ\varphi)(s)](\alpha\circ\varphi)'\mathrm{d}s = \int_{\alpha\circ\varphi} f(\zeta)\mathrm{d}\zeta. \quad \square \end{aligned}$$

Lemma 3.14. Let $f : \mathcal{D} \to \mathbb{C}$ be a continuous function with $\mathcal{D} \subset \mathbb{C}$ open. If f has a primitive F, i.e. $F' = f$, then for any piecewise smooth curve $\alpha \subset \mathcal{D}$ we have

$$\int_\alpha f(\zeta)\mathrm{d}\zeta = F(\alpha(b)) - F(\alpha(a)). \tag{3.13}$$

Proof. Let $F = U + \mathbf{i}V$, then the Cauchy–Riemann equations yield $f = \frac{\partial U}{\partial x} + \mathbf{i}\frac{\partial V}{\partial x} = \frac{\partial V}{\partial y} - \mathbf{i}\frac{\partial U}{\partial y}$. Write $\alpha(t) = x(t) + \mathbf{i}y(t)$ and we have

$$\begin{aligned} f(\alpha(t))\alpha'(t) &= \frac{\partial U}{\partial x}x' + \frac{\partial U}{\partial y}y' + \mathbf{i}\left(\frac{\partial V}{\partial x}x' + \frac{\partial V}{\partial y}y'\right) \\ &= \frac{\mathrm{d}U}{\mathrm{d}t} + \mathbf{i}\frac{\mathrm{d}V}{\mathrm{d}t}. \end{aligned}$$

Then (3.13) follows from integrating the above identity over α and applying (3.11) and the fundamental theorem of calculus (Theorem B.128). □

Corollary 3.15. $\forall z \in \mathbb{C}_-$, $\mathrm{Log}(z) = L(z) := \int_1^z \frac{1}{\zeta}\mathrm{d}\zeta$.

Proof. Corollary 2.26 states that $\mathrm{Log}(z)$ is analytic in $\mathbb{C}_-$ and $(\mathrm{Log}(z))' = \frac{1}{z}$. The rest follows from Lemma 3.14. □

Theorem 3.16. Let $f : \mathcal{D} \to \mathbb{C}$ be a continuous function with $\mathcal{D} \subset \mathbb{C}$ open. If f has a primitive F, i.e. $F' = f$, then for any closed piecewise smooth curve γ in $\mathcal{D}$ we have

$$\oint_\gamma f(\zeta)\mathrm{d}\zeta = 0. \tag{3.14}$$

Proof. Decompose γ into two curves $\alpha, \beta : [0,1] \to \mathbb{C}$ such that $\gamma = \alpha \cup \beta$, $\alpha(0) = \beta(1)$, and $\alpha(1) = \beta(0)$. The proof is completed by $\oint_\gamma = \int_\alpha + \int_\beta$ and Lemma 3.14. □

Corollary 3.17. For a counterclockwise oriented circle $\gamma(t) = r\exp(\mathbf{i}t)$ where $r > 0$ and $t \in [0, 2\pi]$, we have

$$\forall n \in \mathbb{Z}, \quad \oint_\gamma \zeta^n \mathrm{d}\zeta = \begin{cases} 2\pi\mathbf{i} & \text{if } n = -1, \\ 0 & \text{otherwise.} \end{cases} \tag{3.15}$$

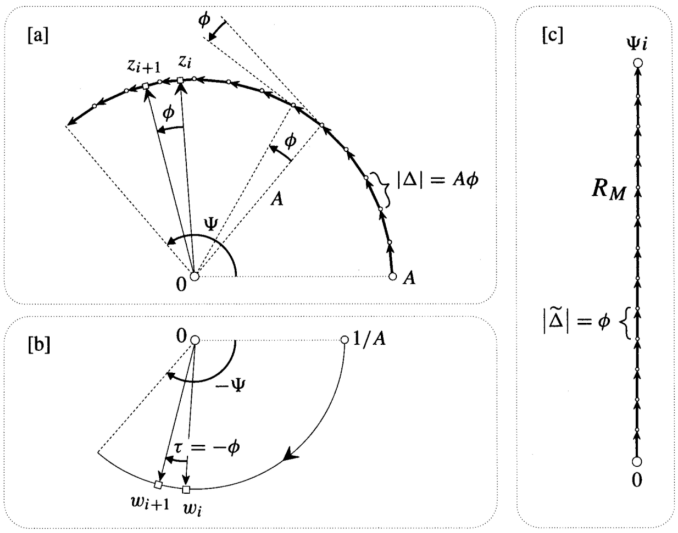
Proof. For $n \neq -1$, the function $f(z) = z^n$ has the primitive $F(z) = \frac{z^{n+1}}{n+1}$, c.f. Exercise 2.22. For $n = -1$,

$$\oint_\gamma \zeta^{-1}\mathrm{d}\zeta = \int_0^{2\pi} (re^{\mathbf{i}t})^{-1} r\mathbf{i}e^{\mathbf{i}t}\mathrm{d}t = 2\pi\mathbf{i}.$$ □

Exercise 3.18 (An alternative proof of Corollary 3.17 via Corollary 3.15)**.** By Corollary 2.26, we have $(\mathrm{Log}z)' = \frac{1}{z}$ for $z \in \mathbb{C}_-$. By Definition 3.24, the unit circle $\mathbb{S}^1 = \alpha \oplus \beta$ is the concatenation of two curves $\theta \mapsto e^{i\theta}$,

$$\alpha := ((-\pi)^+, 0) \to \mathbb{C}, \quad \beta := (0, \pi) \to \mathbb{C}.$$

Put $z_{-1}^- := \alpha((-\pi)^+)$ and $z_{-1}^+ := \beta(\pi)$. Based on the above constructions, starting from the variable substitution $\zeta = rz$, calculate $\oint_\gamma \zeta^{-1}\mathrm{d}\zeta$.



Corollary 3.19. The function $f(z) = \frac{1}{z}$ does not have a primitive if its domain is $\mathbb{C}^\bullet$.

Proof. Otherwise it would contradict Corollary 3.17. □

Exercise 3.20. Draw plots similar to those of $\int_K \frac{1}{z}\mathrm{d}z$ to give a geometric interpretation of $\int_A^B z^m \mathrm{d}z$ where $m \in \mathbb{N}^+$ and the integral is on the arc $\gamma(t) = r\exp(\mathbf{i}t)$ with $t \in [\mathrm{Arg}A, \mathrm{Arg}B]$. Then deduce the formula

$$\int_A^B z^m \mathrm{d}z = \frac{1}{m+1}\left(B^{m+1} - A^{m+1}\right). \tag{3.16}$$

Exercise 3.21. Give a geometric interpretation of $\int_A^B e^z \mathrm{d}z$ and the integral is on the vertical segment $\gamma(t) = A + \mathbf{i}t$ with $t \in [0, \theta]$, where $B = A + \mathbf{i}\theta$ and $\theta > 0$. Then deduce the formula

$$\int_A^B e^z \mathrm{d}z = e^B - e^A. \tag{3.17}$$

Exercise 3.22. In Corollary 3.17, we computed the integral $\oint_\gamma \zeta^n \mathrm{d}\zeta$ for all integers n, where γ is a circle centered at the origin with positive orientation. Without invoking the Cauchy integral theorem, calculate $\oint_\gamma \zeta^n \mathrm{d}\zeta$ where

(a) γ is any circle not enclosing the origin;

(b) γ is a positively oriented square with vertices at $\pm 1, \pm\mathbf{i}$.

Hint: for each of the four line segments, use the parameterization and calculate the real integral $\int_{-1}^1 \frac{1}{1+t^2}\mathrm{d}t$.

Exercise 3.23. For the counterclockwise oriented circle $\gamma(t) = r\exp(\mathbf{i}t)$ with $t \in [0, 2\pi]$ and $r \in (|a|, |b|)$, prove

$$\oint_\gamma \frac{1}{(\zeta - a)(\zeta - b)}\mathrm{d}\zeta = \frac{2\pi\mathbf{i}}{a - b}.$$

Hint: express $\frac{1}{\zeta - a}$ as the geometric series expansion $\frac{1}{1-z} = \sum_{n=0}^\infty z^n$ for $|z| < 1$.

3.3 The Cauchy integral theorem

3.3.1 Simply connected domains

Definition 3.24. The *concatenation of two curves* $\alpha : [a,b] \to \mathbb{C}$ and $\beta : [b,c] \to \mathbb{C}$ with $\alpha(b) = \beta(b)$ is another curve $\alpha \oplus \beta : [a,c] \to \mathbb{C}$ given by

$$(\alpha \oplus \beta)(t) := \begin{cases} \alpha(t) & \text{if } t \in [a,b], \\ \beta(t) & \text{if } t \in [b,c]. \end{cases} \tag{3.18}$$

Example 3.25. $\alpha(t) = \sqrt{1-t^2}$, $t \in [-1,0]$ and $\beta(t) = 1$, $t \in [0,2]$ are two smooth curves. The concatenation $\alpha \oplus \beta$ is not smooth but piecewise smooth: the second derivatives of α and β are different at $t = 0$. Indeed, the curvature of a unit circle and a straight line is 1 and 0, respectively.

Definition 3.26. The *reciprocal curve* of a curve $\alpha : [a,b] \to \mathbb{C}$ is the curve $\alpha^-(t) : [a,b] \to \mathbb{C}$ defined by $\alpha^-(t) = \alpha(b+a-t)$.

Definition 3.27. A set $\mathcal{D} \subset \mathbb{C}$ is called *arcwise connected* or *path connected* iff for any two points $z, w \in \mathcal{D}$ there is a piecewise smooth curve $\alpha : [a,b] \to \mathcal{D}$ such that $\alpha(a) = z$ and $\alpha(b) = w$. In particular, an arcwise connected nonempty open set is called a *domain*.

Example 3.28. The point set $\{z \in \mathbb{C} : |z^2 - 3| < 1\}$ is not a domain whereas, for

$$\mathcal{D} = \{x + \mathbf{i}y : x \in (0,1), y \in (0,1)\};$$
$$\mathcal{D}' = \cup_{n=2}^{\infty} \left\{ x + \mathbf{i}y : x = \frac{1}{n}, y \in \left(0, \frac{1}{2}\right] \right\},$$

$\mathcal{D} \setminus \mathcal{D}'$ is a domain.

Definition 3.29. A *path* in a topological space $\mathcal{X}$ is a continuous map $\zeta : [0,1] \to \mathcal{X}$.

Definition 3.30. A *homotopy of paths* in $\mathcal{X}$ is a family of paths $h_t : [0,1] \to \mathcal{X}$, $t \in [0,1]$, such that $h_t(0)$, $h_t(1)$ are independent of time and the associated map $H : [0,1]^2 \to \mathcal{X}$ defined by $H(s,t) = h_t(s)$ is continuous. Two paths are *homotopic* if they are connected by a homotopy.

Lemma 3.31. The relation of homotopy on paths with fixed endpoints in any space is an equivalence relation.

Definition 3.32. A *loop* or *cycle* is a path whose endpoints coincide; this common endpoint is the *basepoint* of the loop.

Example 3.33. Two loops are homotopic in $\mathcal{X}$ if they are homotopic in $\mathcal{X}$ as paths; in this case, the basepoint of the loop homotopy must be fixed.

Definition 3.34. A *free homotopy* between two loops γ_1 and γ_2 in $\mathcal{X}$ is a function $H_f : [0,1]^2 \to \mathcal{X}$ such that $H_f(s,0) = \gamma_1(s)$ and $H_f(s,1) = \gamma_2(s)$ for all s, and $H_f(0,t) = H_f(1,t)$ for all t. Then γ_1 and γ_2 are said to be *freely homotopic* in $\mathcal{X}$.

Definition 3.35. A topological space $\mathcal{X}$ is *simply connected* if and only if it is path connected and whenever $p : [0,1] \to \mathcal{X}$ and $q : [0,1] \to \mathcal{X}$ are two paths with the same start and end points ($p(0) = q(0)$ and $p(1) = q(1)$), then there exists a homotopy $F : [0,1]^2 \to X$ such that $F(x,0) = p(x)$ and $F(x,1) = q(x)$, i.e., p can be continuously deformed into q while keeping both endpoints fixed.

Exercise 3.36. Prove that Theorem 2.72 holds not only for rectangular domains but also for simply connected domains.

3.3.2 A proof by Riemann

Theorem 3.37. For a continuous function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D}$ a domain, the following statements are equivalent:

(a) f has a primitive.

(b) The integral of f along any piecewise smooth closed curve in $\mathcal{D}$ vanishes.

(c) The integral of f along any piecewise smooth curve in $\mathcal{D}$ depends only on the beginning and end points of the curve.

Proof. By Theorem 3.16, (a) implies (b).

To show (b) implies (c), let $\alpha : [a,b] \to \mathcal{D}$ and $\beta : [c,d] \to \mathcal{D}$ be two curves with the same starting and ending points. WLOG we can assume $b = c$. Otherwise set $\varphi(t) = t + c - b$ in Lemma 3.13 and we have $\int_\beta f(\zeta)\mathrm{d}\zeta = \int_{\beta\circ\varphi} f(\zeta)\mathrm{d}\zeta$. It follows that

$$0 = \oint_{\alpha\oplus\beta^-} f = \int_\alpha f - \int_\beta f.$$

To show (c) implies (a), fix a point $z_* \in \mathcal{D}$ and construct the integral

$$F(z) = \int_{z_*}^{z} f(\zeta)\mathrm{d}\zeta, \tag{3.19}$$

where the integral is along a piecewise smooth curve connecting z_* with z in $\mathcal{D}$. It follows from Lemma 2.18, that we have $F'(z_0) = f(z_0)$ for any given $z_0 \in \mathcal{D}$ if the residual

$$r(z) := F(z) - F(z_0) - f(z_0)(z - z_0)$$

satisfies

$$\lim_{z\to z_0} \frac{r(z)}{|z - z_0|} = 0. \tag{3.20}$$

Since $z_0 \in \mathcal{D}$, there exists an open ball $\mathcal{U}_\rho(z_0)$ that is a subset of $\mathcal{D}$. For all $z \in \mathcal{U}_\rho(z_0)$, we have

$$F(z) = \int_{z_*}^{z_0} f(\zeta)\mathrm{d}\zeta + \int_{z_0}^{z} f(\zeta)\mathrm{d}\zeta = F(z_0) + \int_{z_0}^{z} f(\zeta)\mathrm{d}\zeta,$$

where the last integral can be taken from z_0 to z along the line segment $\sigma : [0,1] \to \mathcal{D}$,

$$\sigma(t) := z_0 + t(z - z_0).$$

Lemma 3.6 yields $\int_\sigma \mathrm{d}\zeta = z - z_0$. Hence we have

$$r(z) = \int_\sigma (f(\zeta) - f(z_0))\mathrm{d}\zeta$$

and the continuity of f gives

$$\forall \epsilon > 0,\ \exists \delta \in (0,\rho),\ \text{s.t. } |z - z_0| < \delta \Rightarrow |f(z) - f(z_0)| < \epsilon.$$

Then Lemma 3.7 yields (3.20). The proof is completed by z_0 being arbitrary. □

Theorem 3.38 (Green)**.** Let α be a counter-clockwise oriented, piecewise smooth, simple closed curve in the plane. Denote by $\mathcal{D}$ the bounded complement of α and $u, v : \mathcal{D} \to \mathbb{R}$ functions with continuous partial derivatives on $\mathcal{D}$. Then

$$\oint u(x,y)\mathrm{d}x + v(x,y)\mathrm{d}y = \iint \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) \mathrm{d}x\mathrm{d}y. \quad (3.21)$$

Lemma 3.39. If $f(z) = u(x,y) + \mathbf{i}v(x,y)$ is continuous along a piecewise smooth curve α, then $f(z)$ is integrable along α and

$$\int_\alpha f(z)\mathrm{d}z = \int_\alpha u\mathrm{d}x - v\mathrm{d}y + \mathbf{i}\int_\alpha v\mathrm{d}x + u\mathrm{d}y. \quad (3.22)$$

Proof. This follows from $\mathrm{d}z = \mathrm{d}x + \mathbf{i}\mathrm{d}y$, Definition 3.4, and the fact that a continuous function is integrable. □

Theorem 3.40 (Cauchy)**.** If $f : \mathcal{D} \to \mathbb{C}$ is analytic in a simply connected domain $\mathcal{D}$ and f' is continuous in $\mathcal{D}$, then

$$\oint_{\partial\mathcal{D}} f(\zeta)\mathrm{d}\zeta = 0. \quad (3.23)$$

Proof. Let $f(z) = u(x,y) + \mathbf{i}v(x,y)$ and $z = x + \mathbf{i}y$. In 1851, Riemann reasoned that

$$\oint_{\partial\mathcal{D}} f(\zeta)\mathrm{d}\zeta = \oint_{\partial\mathcal{D}} u\mathrm{d}x - v\mathrm{d}y + \mathbf{i}\oint_{\partial\mathcal{D}} v\mathrm{d}x + u\mathrm{d}y$$
$$= \iint_{\mathcal{D}} -\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)\mathrm{d}x\mathrm{d}y + \mathbf{i}\iint_{\mathcal{D}}\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right)\mathrm{d}x\mathrm{d}y = 0,$$

where the first step follows from Lemma 3.39, the second step follows from Theorem 3.38, the third step follows from Theorem 2.49. □

Lemma 3.41 (Polynomial growth)**.** For a complex polynomial of degree $n \in \mathbb{N}^+$,

$$P(z) = \sum_{j=0}^{n} a_j z^j, \quad (3.24)$$

where $a_j \in \mathbb{C}$ for all $j = 0, \ldots, n$ and $a_n \neq 0$, we have

$$\frac{1}{2}|a_n||z|^n \leq |P(z)| \leq \frac{3}{2}|a_n||z|^n \quad (3.25)$$

for any z with a sufficiently large modulus $|z| \geq \rho$, e.g.,

$$\rho := \max\left\{1,\ \frac{2}{|a_n|}\sum_{j=0}^{n-1}|a_j|\right\}. \quad (3.26)$$

Proof. Define $Q(z) = P(z) - a_n z^n$. Then the triangle inequality and the condition $|z| \geq \rho$ yield

$$|Q(z)| \leq \left(\sum_{j=0}^{n-1}|a_j|\right)|z|^{n-1} \leq \frac{|a_n|}{2}|z|^n.$$

Apply the triangle inequality, i.e. Corollary 1.66, again to $Q(z) + a_n z^n = P(z)$ and we have

$$|a_n||z|^n - |Q(z)| \leq |P(z)| \leq |a_n||z|^n + |Q(z)|.$$

Combining the two inequalities yields (3.25). □

Theorem 3.42 (Fundamental theorem of Algebra)**.** Any nonconstant complex polynomial (3.24) has a root.

Proof. Suppose $P(z) \neq 0$ for all $z \in \mathbb{C}$. Define $Q(z)$ by $P(z) = zQ(z) + a_0$. Then for any $z \neq 0$ we have

$$\frac{1}{z} = \frac{zQ(z) + a_0}{zP(z)} = \frac{Q(z)}{P(z)} + \frac{a_0}{zP(z)}.$$

By Corollary 3.17 and Theorem 3.40, the integral of $\frac{1}{z}$ along the circle $\alpha(t) = R\exp(\mathbf{i}t)$ with $t \in [0, 2\pi]$ and $R > 0$ yields

$$2\pi\mathbf{i} = \oint_\alpha \frac{a_0}{zP(z)}\mathrm{d}z,$$

where we have used the fact of $P(z) \neq 0$, $P(z)$ and $Q(z)$ being analytic. Then Lemma 3.7 and Lemma 3.41 yield

$$2\pi = \left|\oint_\alpha \frac{a_0}{zP(z)}\mathrm{d}z\right| \leq \oint_\alpha \left|\frac{a_0}{zP(z)}\right|\mathrm{d}z \leq 2\pi R\frac{2|a_0|}{R|a_n|R^n},$$

which cannot hold for any sufficiently large $R > 0$. □

3.3.3 A proof by Goursat

Definition 3.43. The *triangle spanned by* $z_1, z_2, z_3 \in \mathbb{C}$ is the point set

$$\Delta := \left\{\sum_{j=1}^{3} t_j z_j :\ t_j \geq 0,\ \sum_{j=1}^{3} t_j = 1\right\}. \quad (3.27)$$

The *triangular path* $\langle z_1, z_2, z_3\rangle$ is the closed curve

$$\langle z_1, z_2, z_3\rangle = \alpha := \alpha_1 \oplus \alpha_2 \oplus \alpha_3, \quad (3.28)$$

where the edges constituting the triangle boundary are

$$\alpha_1(t) = z_1 + (t-0)(z_2 - z_1),\ t \in [0,1];$$
$$\alpha_2(t) = z_2 + (t-1)(z_3 - z_2),\ t \in [1,2];$$
$$\alpha_3(t) = z_3 + (t-2)(z_1 - z_3),\ t \in [2,3].$$

Theorem 3.44 (Cantor intersection theorem)**.** Suppose $(\mathcal{P}_k)$ is a sequence of non-empty, compact subsets of $\mathcal{P}$ satisfying $\mathcal{P}_0 \supseteq \mathcal{P}_1 \supseteq \cdots \supseteq \mathcal{P}_k \supset \cdots$. Then $\cap_k \mathcal{P}_k \neq 0$.

Exercise 3.45. Prove that if $\{E_n\}$ is a sequence of closed nonempty and bounded sets in a complete metric space $\mathcal{X}$, if $E_{n+1} \subset E_n$, and if $\lim_{n\to\infty} \operatorname{diam} E_n = 0$, cf. Definition D.22. Then $\cap_{n=1}^{\infty} E_n$ consists of exactly one point.

Exercise 3.46. Use the conclusion in Exercise 3.45 to prove the Baire category theorem D.80.

Theorem 3.47 (Cauchy integral theorem on triangular paths)**.** If $f : \mathcal{D} \to \mathbb{C}$ is analytic on a domain $\mathcal{D}$ and the triangle $\Delta(z_1, z_2, z_3)$ is a subset of $\mathcal{D}$, then we have

$$\oint_{\langle z_1, z_2, z_3\rangle} f(\zeta)\mathrm{d}\zeta = 0. \quad (3.29)$$

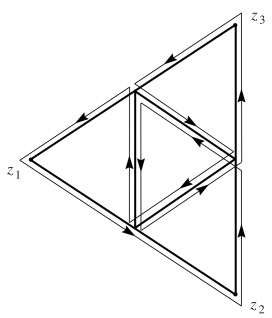
Proof. As shown in the plot below, we first construct a sequence of triangular paths

$$\alpha^{(n)} = \left\langle z_1^{(n)}, z_2^{(n)}, z_3^{(n)} \right\rangle$$

by inductively defining

$$\begin{aligned}
\alpha^{(0)} &:= \alpha = \langle z_1, z_2, z_3 \rangle, \\
\alpha_1^{(n)} &:= \left\langle \frac{z_1^{(n)} + z_2^{(n)}}{2}, z_2^{(n)}, \frac{z_2^{(n)} + z_3^{(n)}}{2} \right\rangle, \\
\alpha_2^{(n)} &:= \left\langle \frac{z_2^{(n)} + z_3^{(n)}}{2}, z_3^{(n)}, \frac{z_3^{(n)} + z_1^{(n)}}{2} \right\rangle, \\
\alpha_3^{(n)} &:= \left\langle \frac{z_3^{(n)} + z_1^{(n)}}{2}, z_1^{(n)}, \frac{z_1^{(n)} + z_2^{(n)}}{2} \right\rangle, \\
\alpha_4^{(n)} &:= \left\langle \frac{z_1^{(n)} + z_2^{(n)}}{2}, \frac{z_2^{(n)} + z_3^{(n)}}{2}, \frac{z_1^{(n)} + z_3^{(n)}}{2} \right\rangle.
\end{aligned}$$

and choosing $\alpha^{(n+1)} = \alpha_1^{(n)}$ or $\alpha_2^{(n)}$ or $\alpha_3^{(n)}$ or $\alpha_4^{(n)}$.



It follows that

$$\oint_{\alpha^{(n)}} = \oint_{\alpha_1^{(n)}} + \oint_{\alpha_2^{(n)}} + \oint_{\alpha_3^{(n)}} + \oint_{\alpha_4^{(n)}}$$

and we can choose $\alpha^{(n+1)}$ such that

$$\left| \oint_{\alpha^{(n)}} f \right| \le 4 \left| \oint_{\alpha^{(n+1)}} f \right|,$$

which implies

$$\left| \oint_{\alpha} f \right| \le 4^n \left| \oint_{\alpha^{(n)}} f \right|.$$

The closed triangles $\Delta^{(n)}$ are nested and Theorem 3.44 implies that there exists a point $z_0 \in \cap_n \Delta^{(n)} \subset \mathcal{D}$. By Lemma 2.18, the condition of f being analytic at z_0 yields $f(z) = f(z_0) + f'(z_0)(z - z_0) + r(z)$ where

$$\lim_{z \to z_0} \frac{r(z)}{|z - z_0|} = 0. \tag{3.30}$$

Then we have $\oint_{\alpha^{(n)}} f(\zeta)\mathrm{d}\zeta = \oint_{\alpha^{(n)}} r(\zeta)\mathrm{d}\zeta$ from Theorem 3.16 and Exercise 2.22. It follows that

$$\left| \oint_{\alpha} f(\zeta)\mathrm{d}\zeta \right| \le 4^n \left| \oint_{\alpha^{(n)}} r(\zeta)\mathrm{d}\zeta \right|. \tag{3.31}$$

We show $\lim_{n\to\infty} 4^n \left| \oint_{\alpha^{(n)}} r(\zeta)\mathrm{d}\zeta \right| = 0$ to complete the proof. By (3.30), we have

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } |z - z_0| < \delta \Rightarrow |r(z)| \le \epsilon |z - z_0|, \tag{3.32}$$

where the condition $|z - z_0| < \delta$ holds for any sufficiently large n. In other words, we can choose n large enough such that $\Delta^{(n)} \subset \mathcal{U}_\delta(z_0)$. Denote by $\ell(\alpha)$ the arc length of α as in Definition C.382. Then by construction of the sequence of triangles, we have

$$|z - z_0| \le \ell(\alpha^{(n)}) = \frac{1}{2^n}\ell(\alpha),$$

which, together with (3.31), (3.32), and Lemma 3.7, yields

$$\left| \oint_{\alpha} f(\zeta)\mathrm{d}\zeta \right| \le 4^n \epsilon \frac{1}{2^n} \ell(\alpha)\ell(\alpha^{(n)}) = \epsilon \ell^2(\alpha).$$

To sum up, $\left| \oint_\alpha f(\zeta)\mathrm{d}\zeta \right|$ can be made arbitrarily small, hence (3.29) holds. □

Corollary 3.48. If $f : \mathcal{D} \to \mathbb{C}$ is analytic in a simply connected domain $\mathcal{D}$, then for any piecewise linear closed curve $\beta \subset \mathcal{D}$ we have

$$\oint_\beta f(\zeta)\mathrm{d}\zeta = 0. \tag{3.33}$$

Proof. This follows from Theorem 3.47, Lemma 8.6, and the two ears theorem, which states that every simple polygon with more than three vertices has at least two ears. □

Theorem 3.49 (Heine 1872)**.** If $K \subset \mathbb{C}$ is compact and $f : K \to \mathbb{C}$ is continuous, then f is uniformly continuous.

Proof. This is a specialization of Theorem D.150. □

Lemma 3.50. If $f : \mathcal{D} \to \mathbb{C}$ is continuous in a domain $\mathcal{D}$, then for any piecewise smooth simple curve $\alpha \subset \mathcal{D}$ and any $\epsilon > 0$, there exists a piecewise linear curve $\sigma \subset \mathcal{D}$ such that

$$\left| \int_\alpha f(\zeta)\mathrm{d}\zeta - \int_\sigma f(\zeta)\mathrm{d}\zeta \right| < \epsilon. \tag{3.34}$$

In other words, the integral of f on a piecewise smooth curve can be approximated arbitrarily well by an integral of f on line segments.

Proof. Since $\mathcal{D}$ is open, there exists a compact subset $G \subset \mathcal{D}$ such that $\alpha \cap \partial G = \emptyset$ and $\alpha \subset G$. Define

$$\rho = \inf\{|z_1 - z_2| : z_1 \in \alpha, z_2 \in \partial G\}.$$

Clearly $\rho > 0$. We have $\forall z \in \alpha, \forall r < \rho,\ \mathcal{U}_r(z) \subset G$ and

$$\forall z_1, z_2 \in \alpha,\ |z_1 - z_2| < \rho \Rightarrow \{z_1 t + z_2(1-t) : t \in [0,1]\} \subset G.$$

By Theorem 3.49, the continuity of f and the compactness of G yield the uniform continuity of f on G, i.e.

$$\begin{aligned}
&\forall \epsilon > 0, \exists \delta_1 > 0 \text{ s.t. } \forall z_1, z_2 \in G, \\
&\qquad |z_1 - z_2| < \delta_1 \Rightarrow |f(z_1) - f(z_2)| < \frac{\epsilon}{2\ell},
\end{aligned} \tag{3.35}$$

where ℓ is the arc length of α.

Choose $n+1$ points $z_1, z_2, \ldots, z_{n+1} \in \alpha$ such that $\max_{j=1}^n |z_j - z_{j+1}| < \min(\rho, \delta_1)$: if α is closed then we set $z_{n+1} = z_1$; if α is not closed we set z_1 and z_{n+1} as the starting and ending points, respectively. We also require

that $\alpha^{-1}(z_1)$, $\alpha^{-1}(z_2)$, …, $\alpha^{-1}(z_{n+1})$ increases monotonically. Then the piecewise linear closed curve σ formed by connecting z_j's consecutively by linear segments is contained in $\mathcal{D}$. Its edges σ_j's correspond to the arcs α_j's. For each j, Lemma 3.14 yields

$$\int_{\sigma_j} f(z_j)\mathrm{d}\zeta = f(z_j)(z_{j+1} - z_j) = \int_{\alpha_j} f(z_j)\mathrm{d}\zeta. \tag{3.36}$$

Consequently, we have

$$\begin{aligned}
&\left|\int_{\alpha_j} f(\zeta)\mathrm{d}\zeta - \int_{\sigma_j} f(\zeta)\mathrm{d}\zeta\right| \\
\le&\left|\int_{\alpha_j} [f(\zeta) - f(z_j)]\,\mathrm{d}\zeta\right| + \left|\int_{\sigma_j} [f(\zeta) - f(z_j)]\,\mathrm{d}\zeta\right| \\
\le&\ell(\alpha_j) \sup_{\zeta\in\alpha_j} |f(\zeta) - f(z_j)| + \ell(\sigma_j) \sup_{\zeta\in\sigma_j} |f(\zeta) - f(z_j)| \\
<&\frac{\epsilon}{2\ell}\left[\ell(\alpha_j) + \ell(\sigma_j)\right] \le \frac{\epsilon}{\ell}\ell(\alpha_j),
\end{aligned}$$

where the first step follows from (3.36) and the triangle inequality, the second step from Lemma 3.7 and Definition C.382, and the third step from (3.35). Then we have

$$\begin{aligned}
\left|\int_{\alpha} f(\zeta)\mathrm{d}\zeta - \int_{\sigma} f(\zeta)\mathrm{d}\zeta\right| =& \left|\sum_{j=1}^{n}\int_{\alpha_j} f(\zeta)\mathrm{d}\zeta - \int_{\sigma_j} f(\zeta)\mathrm{d}\zeta\right| \\
\le& \sum_{j=1}^{n}\left|\int_{\alpha_j} f(\zeta)\mathrm{d}\zeta - \int_{\sigma_j} f(\zeta)\mathrm{d}\zeta\right| \\
=&\epsilon,
\end{aligned}$$

which completes the proof. □

Theorem 3.51 (Cauchy integral theorem for simply connected domains)**.** If $f : \mathcal{D} \to \mathbb{C}$ is analytic in a simply connected domain $\mathcal{D}$, then for any piecewise smooth closed curve $\alpha \subset \mathcal{D}$ we have

$$\oint_{\alpha} f(\zeta)\mathrm{d}\zeta = 0. \tag{3.37}$$

Proof. This follows directly from Corollary 3.48 and Lemma 3.50. □

Definition 3.52. A *star-shaped domain* is an open set $\mathcal{D} \subset \mathbb{C}$ that satisfies

$$\exists z_* \in \mathcal{D} \text{ s.t. } \forall z \in \mathcal{D},\ \{z_* + t(z - z_*) : t \in [0, 1]\} \subset \mathcal{D},$$

where z_* is called a *star center*.

Example 3.53. Each convex domain, in particular the open disk, is star-shaped. The ring sector

$$\left\{z_0 + \zeta\rho e^{\mathbf{i}\varphi} :\ \rho \in (r, R), \varphi \in (0, \beta);\ \zeta, z_0 \in \mathbb{C}, |\zeta| = 1\right\}$$

is star-shaped if $\beta < \pi$ and $\cos\frac{\beta}{2} > \frac{r}{R}$.

Theorem 3.54 (Cauchy integral theorem for star-shaped domains)**.** Each analytic function f defined on a star-shaped domain $\mathcal{D}$ has a primitive in $\mathcal{D}$.

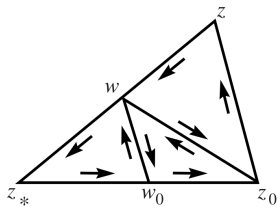
Proof. This follows from Theorems 3.51 and 3.37 since a star-shaped domain is a simply connected domain. □

Corollary 3.55. Let $f : \mathcal{D} \to \mathbb{C}$ be continuous in a star-shaped domain $\mathcal{D}$ with center z_*. If f is analytic at every point $z \neq z_*$, then f has a primitive in $\mathcal{D}$.

Proof. By Theorem 3.37, it suffices to show

$$\oint_{\langle z, z_0, z_*\rangle} f(\zeta)\mathrm{d}\zeta = 0,$$

where $z_* \neq z$, $z_* \neq z_0$, and the triangle is entirely contained within $\mathcal{D}$.



As shown in the above plot, Theorem 3.47 yields

$$\oint_{\langle z_*, z_0, z\rangle} = \oint_{\langle z_*, w_0, w\rangle} + \oint_{\langle w_0, z_0, w\rangle} + \oint_{\langle z_0, z, w\rangle} = \oint_{\langle z_*, w_0, w\rangle}.$$

The proof is completed by passing to the limit $w \to z_*$ and $w_0 \to z_*$. □

3.3.4 Non-simply connected domains

Definition 3.56. A *disk with n holes* ($n \in \mathbb{N}$) is a connected open set $\mathcal{D} \subset \mathbb{C}$ such that it can be expressed as

$$\mathcal{D} := \bigcap_{j=0}^{n} \mathrm{int}(\alpha_j), \tag{3.38}$$

where $\mathrm{int}(\alpha)$ is the interior of a Jordan curve α in Definition 8.5, α_0 is a counter-clockwise oriented Jordan curve, and α_1, …, α_n are clockwise oriented Jordan curves satisfying

$$\forall j = 1, \ldots, n,\ \mathrm{ext}(\alpha_j) \subset \mathrm{int}(\alpha_0), \tag{3.39a}$$

$$\forall j, k = 1, \ldots, n,\ j \neq k\ \Rightarrow \mathrm{ext}(\alpha_j) \cap \mathrm{ext}(\alpha_k) = \emptyset. \tag{3.39b}$$

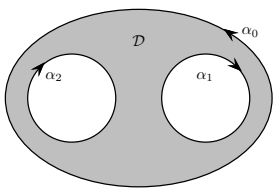
The boundary and closure of a disk with holes are denoted respectively by

$$\partial\mathcal{D} := \cup_{j=0}^{n}\alpha_j, \qquad \overline{\mathcal{D}} := \mathcal{D} \cup \partial\mathcal{D}.$$

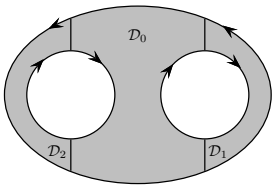
Theorem 3.57 (Cauchy integral theorem for a disk with holes)**.** Let $\mathcal{D} := \cap_{j=0}^{n}\mathrm{int}(\alpha_j)$ be a disk with holes as in Definition 3.56. If a complex function f is analytic on $\mathcal{D}$ and continuous on $\overline{\mathcal{D}} := \mathcal{D} \cup \partial\mathcal{D}$, then we have

$$\int_{\partial\mathcal{D}} f(z)\mathrm{d}z = 0.$$

Proof. The domain $\mathcal{D}$ is exemplified for $n = 2$ as follows.



It follows from a simple induction that $\cup_{j=0}^n \alpha_j$ can be reorganized into positively-oriented Jordan curves with disjoint interiors $\mathcal{D}_0, \mathcal{D}_1, \ldots, \mathcal{D}_m$ as follows.



The proof is then completed by $\cup_{j=0}^n \alpha_j = \cup_{j=0}^m \partial\mathcal{D}_j$ and the Cauchy integral theorem 3.51. □

3.4 The Cauchy integral formulas

Lemma 3.58. Define a circle as

$$\alpha(t) = z_0 + re^{\mathbf{i}t},\ z_0 \in \mathbb{C}, t \in [0, 2\pi], r > 0. \tag{3.40}$$

Then we have

$$\forall a \in \mathbb{C},\ |a - z_0| < r \ \Rightarrow\ \oint_\alpha \frac{\mathrm{d}\zeta}{\zeta - a} = 2\pi\mathbf{i}. \tag{3.41}$$

Proof. Construct a disk $\mathcal{D}$ with one hole by setting α_0 as α and α_1 as a clockwise oriented circle of which the center is a and the radius is $\rho \le r - |z_0 - a|$. Theorem 3.57 yields

$$0 = \int_{\partial\mathcal{D}} = \oint_\alpha + \oint_{\alpha_1} \implies \oint_\alpha = \oint_{\alpha_1^-}.$$

The rest of the proof follows from Corollary 3.17. □

Theorem 3.59 (Cauchy integral formula)**.** Let $\mathcal{D} \subseteq \mathbb{C}$ be open and $f : \mathcal{D} \to \mathbb{C}$ be an analytic function. Assume that the closed disk $\overline{\mathcal{U}}_r(z_0)$ is contained completely in $\mathcal{D}$. Then we have

$$\forall z \in \mathcal{U}_r(z_0), \qquad f(z) = \frac{1}{2\pi\mathbf{i}} \oint_\alpha \frac{f(\zeta)}{\zeta - z}\mathrm{d}\zeta, \tag{3.42}$$

where α is the circle defined in (3.40).

Proof. By the compactness of $\overline{\mathcal{U}}_r(z_0)$, there exists an $R > r$ such that

$$\overline{\mathcal{U}}_r(z_0) \subset \mathcal{U}_R(z_0) \subset \mathcal{D}.$$

We can thus assume that $\mathcal{D}$ is an open disk. The function

$$g(w) := \begin{cases} \frac{f(w)-f(z)}{w-z} & \text{if } w \ne z, \\ f'(z) & \text{if } w = z \end{cases}$$

is analytic in $\mathcal{D}\backslash\{z\}$ and is continuous in $\mathcal{D}$. Then Corollary 3.55 gives

$$\oint \frac{f(\zeta) - f(z)}{\zeta - z}\mathrm{d}\zeta = 0 \implies f(z) \oint \frac{1}{\zeta - z}\mathrm{d}\zeta = \oint \frac{f(\zeta)}{\zeta - z}\mathrm{d}\zeta.$$

The proof is completed by Lemma 3.58. □

Corollary 3.60 (Cauchy integral formula for non-simply connected domains)**.** Let α_0 be a counter-clockwise oriented Jordan curve and $\alpha_1, \ldots, \alpha_n$ be clockwise oriented Jordan curves satisfying (3.39). If a complex function f is analytic on the disk with holes $\mathcal{D} := \cap_{j=0}^n \mathrm{int}(\alpha_j)$ and continuous on $\overline{\mathcal{D}}$, then we have

$$\oint_{\alpha_0} f(z)\mathrm{d}z = \sum_{j=1}^n \oint_{\alpha_j^+} f(z)\mathrm{d}z, \tag{3.43}$$

where each α_j^+ is α_j with its orientation reversed.

Proof. This follows from Theorem 3.57 by moving the summation term in (3.43) to the right-hand side (RHS) and reversing the orientation of α_j's for each $j = 1, 2, \ldots, n$. □

Example 3.61. Let α denote the circle defined in (3.40) with $z_0 = 0$ and $r = 2$. We compute

$$\begin{aligned} \oint_\alpha \frac{\zeta}{(9-\zeta^2)(\zeta+\mathbf{i})}\mathrm{d}\zeta &= \oint_\alpha \frac{\frac{\zeta}{9-\zeta^2}}{\zeta - (-\mathbf{i})}\mathrm{d}\zeta \\ &= 2\pi\mathbf{i}\ \frac{\zeta}{9-\zeta^2}\bigg|_{\zeta=-\mathbf{i}} = \frac{\pi}{5} \end{aligned}$$

where the second step follows from Theorem 3.59.

Corollary 3.62 (Mean value equation)**.** For the special case of $z = z_0$, Theorem 3.59 reduces to

$$f(z_0) = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + r\exp(\mathbf{i}t))\mathrm{d}t. \tag{3.44}$$

Proof. This follows directly from Theorem 3.59 by substituting $z = z_0$ and $\zeta = z_0 + re^{\mathbf{i}t}$. □

Example 3.63. Compute $I = \oint_{|z|=2} \frac{\mathrm{d}z}{z^2+2}$.

Construct two counter-clockwise oriented circles α_1 and α_2 surrounding the two singularities $z = \pm\sqrt{2}\mathbf{i}$. For sufficiently small radii, Corollary 3.60 yields

$$\begin{aligned} \oint_{|z|=2} f(z)\mathrm{d}z &= \oint_{\alpha_1} f(z)\mathrm{d}z + \oint_{\alpha_2} f(z)\mathrm{d}z \\ &= \oint_{\alpha_1} \frac{\frac{1}{z+\sqrt{2}\mathbf{i}}}{z - \sqrt{2}\mathbf{i}}\mathrm{d}z + \oint_{\alpha_2} \frac{\frac{1}{z-\sqrt{2}\mathbf{i}}}{z - (-\sqrt{2}\mathbf{i})}\mathrm{d}z \\ &= \frac{2\pi\mathbf{i}}{z+\sqrt{2}\mathbf{i}}\bigg|_{z=\sqrt{2}\mathbf{i}} + \frac{2\pi\mathbf{i}}{z-\sqrt{2}\mathbf{i}}\bigg|_{z=-\sqrt{2}\mathbf{i}} \\ &= 0. \end{aligned}$$

Theorem 3.64 (Leibniz integral rule)**.** Let $f : [a, b] \times [c, d] \to \mathbb{R}$ be a continuous function. Suppose that the partial derivative $\frac{\partial f(t,x)}{\partial x}$ exists and is continuous. Also suppose that the functions $\alpha, \beta : [c, d] \to [a, b]$ are both continuously differentiable. Then

$$\begin{aligned} &\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{\alpha(x)}^{\beta(x)} f(t, x)\mathrm{d}t\right) \\ &= \int_{\alpha(x)}^{\beta(x)} \frac{\partial f(t,x)}{\partial x}\mathrm{d}t + f(\beta(x), x)\frac{\mathrm{d}\beta(x)}{\mathrm{d}x} - f(\alpha(x), x)\frac{\mathrm{d}\alpha(x)}{\mathrm{d}x}. \end{aligned} \tag{3.45}$$

Proof. This follows from the chain rule. □

Lemma 3.65. Let $\mathcal{D} \subseteq \mathbb{C}$ be open and $f : [a,b] \times \mathcal{D} \to \mathbb{C}$ be a continuous function that is also analytic in $\mathcal{D}$ for any fixed $t \in [a,b]$. Also assume that the partial derivative $\frac{\partial f}{\partial z} : [a,b] \times \mathcal{D} \to \mathbb{C}$ is continuous. Then the function

$$g(z) := \int_a^b f(t,z)\mathrm{d}t$$

is analytic in $\mathcal{D}$ and

$$g'(z) = \int_a^b \frac{\partial f(t,z)}{\partial z}\mathrm{d}t. \tag{3.46}$$

Proof. For any fixed t, write $f(t,z) = p(t,x,y) + \mathbf{i}q(t,x,y)$ and $g(z) = u(x,y) + \mathbf{i}v(x,y)$. Then we have $u = \int p\mathrm{d}t$ and $v = \int q\mathrm{d}t$. Then the Cauchy–Riemann theorem 2.49 applied to f yields

$$\begin{aligned}\frac{\partial u}{\partial x} &= \int \frac{\partial p}{\partial x} = \int \frac{\partial q}{\partial y} = \frac{\partial v}{\partial y},\\ \frac{\partial u}{\partial y} &= \int \frac{\partial p}{\partial y} = -\int \frac{\partial q}{\partial x} = -\frac{\partial v}{\partial x}.\end{aligned}$$

Since $\frac{\partial f}{\partial z} : [a,b] \times \mathcal{D} \to \mathbb{C}$ is continuous, $g(z)$ is totally differentiable and thus analytic by the above equations and the Cauchy–Riemann theorem. The formula (3.46) follows from Theorem 3.64. □

Theorem 3.66 (Generalized Cauchy integral formula)**.** Every analytic function is arbitrarily often complex differentiable. Each derivative is again analytic. Furthermore,

$$f^{(n)}(z) = \frac{n!}{2\pi\mathbf{i}} \oint_\alpha \frac{f(\zeta)}{(\zeta - z)^{n+1}}\mathrm{d}\zeta, \tag{3.47}$$

where the assumptions and notation are the same as those in Theorem 3.59.

Proof. Define

$$G(t,z) := \frac{f(\zeta(t))\zeta'(t)}{\zeta(t) - z}.$$

From Theorem 3.59 we have

$$f(z) = \frac{1}{2\pi\mathbf{i}} \oint_\alpha G(t,z)\mathrm{d}t.$$

The rest of the proof follows from Lemma 3.65 and an induction on n. □

Corollary 3.67 (Cauchy inequality)**.** If $f : \mathcal{D} \to \mathbb{C}$ is analytic and $\mathcal{U}_r(z_0) \subset \mathcal{D}$ for some $r > 0$, then we have

$$\forall n \in \mathbb{N}^+, \qquad |f^{(n)}(z_0)| \le \frac{n!M(r)}{r^n}, \tag{3.48}$$

where $M(r) = \max_{|z-z_0|=r} |f(z)|$.

Proof. For the circle defined in (3.40), Theorem 3.66 yields

$$\begin{aligned}|f^{(n)}(z_0)| &= \left|\frac{n!}{2\pi\mathbf{i}} \oint_\alpha \frac{f(\zeta)}{(\zeta - a)^{n+1}}\mathrm{d}\zeta\right|\\ &\le \frac{n!}{2\pi}\frac{M(r)}{r^{n+1}}2\pi r = \frac{n!M(r)}{r^n}.\end{aligned}$$ □

Example 3.68. For a Jordan curve α around $\mathbf{i}$, we have

$$\begin{aligned}\oint_\alpha \frac{\cos z}{(z-\mathbf{i})^3}\mathrm{d}z &= \frac{2\pi\mathbf{i}}{2!}\left.(\cos z)''\right|_{z=\mathbf{i}} = -\pi\mathbf{i}\cos\mathbf{i}\\ &= -\pi\frac{e^{-1}+e}{2}\mathbf{i}.\end{aligned}$$

3.5 Morera's and Liouville's theorems

Theorem 3.69 (Morera 1886)**.** Let $\mathcal{D} \subseteq \mathbb{C}$ be open and $f : \mathcal{D} \to \mathbb{C}$ be a continuous function. If

$$\oint_{\langle z_1,z_2,z_3\rangle} f(\zeta)\mathrm{d}\zeta = 0 \tag{3.49}$$

for every triangular path $\langle z_1, z_2, z_3\rangle$ whose triangle is entirely contained in $\mathcal{D}$, then f is analytic.

Proof. For any given point $z_0 \in \mathcal{D}$, there is an open neighborhood $\mathcal{U}_\epsilon(z_0) \subset \Delta(z_1,z_2,z_3) \subset \mathcal{D}$. It suffices to show that f is analytic in $\mathcal{U}_\epsilon(z_0)$. By (3.49), Lemma 3.50, and the fact that every polygon has a triangulation, the integral of f along any closed curve in $\mathcal{U}_\epsilon(z_0)$ vanishes. Theorem 3.37 then implies that f has a primitive F and the proof is completed by applying Theorem 3.66 to F. □

Definition 3.70. An analytic function $f : \mathbb{C} \to \mathbb{C}$ is said to be *entire*.

Theorem 3.71 (Liouville 1847)**.** Every bounded entire function is a constant.

Proof. We show $f'(z) = 0$ for all $z \in \mathbb{C}$. The generalized Cauchy integral formula (3.47) states that

$$\forall z \in \mathbb{C}, \forall r > 0, \qquad f'(z) = \frac{1}{2\pi\mathbf{i}} \oint_{|\zeta-z|=r} \frac{f(\zeta)}{(\zeta - z)^2}\mathrm{d}\zeta.$$

Define $C = \max_{|\zeta-z|=r} |f(\zeta)|$ and we have $C < +\infty$ since f is bounded. Then Lemma 3.7 yields

$$|f'(z)| \le \frac{1}{2\pi}2\pi r\frac{C}{r^2} = \frac{C}{r}.$$

Passing to the limit $r \to +\infty$ completes the proof. □

Example 3.72. $f(z) = \cos(z)$ is not bounded since

$$\lim_{x\to+\infty} \cos(\mathbf{i}x) = \lim_{x\to+\infty} \frac{e^x + e^{-x}}{2} = +\infty.$$